UNIQUENESS OF THE MIDDLE ARROW OF A MORPHISM OF TRIANGLES UNDER VANISHING CONDITIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 794-830, following the statement at 775-792.

Editorial note. No omissions for (1), (3), (5); conditions (2) and (4) are explicitly marked dual to (1), following the established dual-mirror counting practice.

Allowed standard prerequisites (editorial).

- Representable functor is homological. lemma-representable-homological (same chapter, used throughout, first cited at ~786). Long exact Hom sequences behind every lifting step; the author uses it without further mention.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category. Let*

$$(a, b, c) : (X, Y, Z, f, g, h) \to (X', Y', Z', f', g', h')$$

be a morphism of distinguished triangles. If one of the following conditions holds

(1) $\mathrm{Hom}(Y, X') = 0$,
(2) $\mathrm{Hom}(Z, Y') = 0$,
(3) $\mathrm{Hom}(X, X') = \mathrm{Hom}(Z, X') = 0$,
(4) $\mathrm{Hom}(Z, X') = \mathrm{Hom}(Z, Z') = 0$, *or*
(5) $\mathrm{Hom}(X[1], Z') = \mathrm{Hom}(Z, X') = 0$

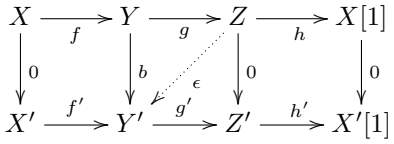
then b is the unique morphism from $Y \to Y'$ such that (a, b, c) is a morphism of triangles.

Proof. If we have a second morphism of triangles (a, b', c) then $(0, b - b', 0)$ is a morphism of triangles. Hence we have to show: the only morphism $b : Y \to Y'$ such that $X \to Y \to Y'$ and $Y \to Y' \to Z'$ are zero is 0. We will use Lemma 0149 without further mention. In particular, condition (3) implies (1). Given condition (1) if the composition $g' \circ b : Y \to Y' \to Z'$ is zero, then b lifts to a morphism $Y \to X'$ which has to be zero. This proves (1).

The proof of (2) and (4) are dual to this argument.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Assume (5). Consider the diagram



We may choose ϵ such that $b = \epsilon \circ g$. Then $g' \circ \epsilon \circ g = 0$ which implies that $g' \circ \epsilon = \delta \circ h$ for some $\delta \in \mathrm{Hom}(X[1], Z')$. Since $\mathrm{Hom}(X[1], Z') = 0$ we conclude that $g' \circ \epsilon = 0$. Hence $\epsilon = f' \circ \gamma$ for some $\gamma \in \mathrm{Hom}(Z, X')$. Since $\mathrm{Hom}(Z, X') = 0$ we conclude that $\epsilon = 0$ and hence $b = 0$ as desired. $\square$